MIRPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY

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SUBJECT:

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DATE:

26 APRIL 2026

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REPORT TYPE:

LAB TASK

DEPARTMENT:

SOFTWARE ENGINEERING

QUALITY ASSESSMENT REPORT

ISO 25010 Comparative Analysis: Chrome Mobile vs Safari

Evaluator: ABDUR REHMAN
Student ID: FA23-BSE-003
Section/Group: A
Date of Evaluation: April 21, 2026
Pair Selected: D — Chrome Mobile vs Safari

Section 1: Cover Information

Field	App 1	App 2
App Name	Google Chrome Mobile	Apple Safari
Version Number	147.0.7727.101	26
Platform	Android	iOS
OS Version	Android 13.5.0	iOS 26.4.1
Device Model	Tecno Spark 20 Pro	iPhone 17 Pro Max
Min OS Required	Android 10	iOS 10
App Store Rating	4.4/5.0	4.2/5.0
Last Updated	April 15, 2026	April 15, 2026

Section 2: App Overviews

Google Chrome Mobile (Version 147.0.7727.101)

Google Chrome is a cross-platform web browser developed by Google, serving as the default browser on Android devices and one of the most popular browsers worldwide. Its primary purpose is to provide fast, secure, and feature-rich web browsing across desktop and mobile platforms. Chrome targets a broad user base ranging from casual internet users to power users who require seamless synchronization across multiple devices. Key features include tabbed

browsing, incognito mode for private browsing, cross-device sync via Google account, built-in password manager, extensive extension support (desktop), and robust developer tools. The app falls under the productivity category and is available on Android, iOS, Windows, macOS, Linux, and Web platforms. A notable characteristic of Chrome is its emphasis on cross-platform availability and frequent update cycles.

Apple Safari (Version 26)

Safari is Apple's native web browser, exclusively available on iOS and macOS devices. It serves as the default browser for the entire Apple ecosystem and is deeply integrated with Apple's services including iCloud Keychain, Handoff, and Continuity. Safari targets Apple device users who prioritize system integration, privacy, and performance optimization within the Apple ecosystem. Key features include tab groups, private browsing mode, iCloud tab synchronization, built-in privacy protections (Intelligent Tracking Prevention), and seamless integration with Apple's authentication systems (Face ID, Touch ID). The app belongs to the productivity category but differs fundamentally from Chrome in its platform exclusivity — Safari is designed exclusively for Apple hardware and software environments, prioritizing optimization over cross-platform reach.

Section 3: Evaluation

This evaluation was conducted following the ISO/IEC 25010 quality model, assessing eight quality characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Rating Scale: A 5-point Likert scale was used where 5 = Excellent (exceeds expectations), 4 = Good (mostly satisfied with minor issues), 3 = Acceptable (noticeable but usable), 2 = Poor (significant issues), and 1 = Unacceptable (critical failures).

Testing Environment:

- **Device 1 (Chrome):** Tecno Spark 20 Pro running Android 13.5.0
- **Device 2 (Safari):** iPhone 17 Pro Max running iOS 26.4.1
- **Network:** Campus WiFi (primary), 4G cellular for slow network testing
- **Duration:** Approximately 50 minutes per app
- **Evaluation Date:** April 21, 2026

Testing Tools & Methods:

- Stopwatch/timer for cold start times and operation response times
- Device battery settings for power consumption measurement
- Device data settings for data usage monitoring
- Airplane Mode toggle for offline/recovery testing
- Screenshot capture for evidence documentation
- Usability tasks: 5 unguided timed tasks per app

Testing Conditions: Both apps were evaluated under identical network conditions, on the same day, with consistent testing protocols. Each characteristic required a minimum of 2

specific observations with actual measurements where applicable (timings in seconds, percentage data, specific feature names).

Section 4: Completed Checklists

App 1: Google Chrome Mobile — All 8 ISO 25010 Characteristics

Quality Characteristic	Rating (1-5)	Weight	Observations / Evidence
1. Functional Suitability	4.5	15%	**Observation 1:** Primary core function passed — opened a website normally without any issues. All tested features (bookmarks, incognito mode) function correctly. **Observation 2:** Error handling is robust — Chrome displays security warnings for non-HTTPS websites and shows "no internet connection" message when offline. Edge case with 15-20 tabs caused lag and slow switching, indicating partial performance under extreme load.
2. Performance Efficiency	4	12%	**Observation 1:** Cold start time measured at 2.4 seconds (target <3 seconds). First website loaded in 1.3 seconds. Image loading took 2 seconds. **Observation 2:** On slow 4G network, website loaded in 10.25 seconds (exceeds <5 second target). Battery drain was excellent at 1% over 10 minutes. Data usage was 5MB in 10 minutes (under 10MB target).
3. Compatibility	5	10%	**Observation 1:** Chrome is available on all platforms (Android, iOS, Windows, macOS, Web, Linux). Cross-device sync works seamlessly — browsing history from laptop appears on mobile. **Observation 2:** Supports dark mode that respects OS settings. Content sharing works both directions — can receive shares from other apps and share downloaded files to 3+ apps. No features broken on Android 13.5.0.
4. Usability	4.5	15%	**Observation 1:** Usability tasks completed without errors: change core setting (10 sec), primary function first time (2 sec), find help section (5 sec), share

			content (15 sec), delete item (5 sec). **Observation 2:** Error messages are clear and actionable. Visual hierarchy effectively prioritizes important information. Accessibility supports large text and screen readers. No onboarding tutorial for new users.
5. Reliability	4.5	15%	**Observation 1:** No crashes during 10-minute continuous use. Airplane Mode test showed graceful handling with "no internet connection" message. Reconnect recovery successfully synced data after connectivity restored. **Observation 2:** Force-close mid-task partially saves progress — tabs are restored but reload content. Server error messages are meaningful ("Server Not found").
6. Security	4	15%	**Observation 1:** Uses HTTPS with padlock indicator. Permissions are justified for app purpose. 2FA/biometric login available in settings (Google account integration). **Observation 2:** Privacy policy clearly documents data collection practices. Login lockout activates after 3 incorrect attempts. Missing auto-lock on inactivity — app stays open indefinitely.
7. Maintainability	4	8%	**Observation 1:** Updates released weekly on Play Store. Changelog clearly documents bug fixes with specific details. **Observation 2:** No widespread long-standing unfixed bugs in app store reviews. Public bug tracker/feedback channel exists for user reporting.
8. Portability	5	10%	**Observation 1:** Available on all major platforms (Android, iOS, Windows, macOS, Web, Linux). Installation and uninstallation process is clean with no residual files observed. **Observation 2:** Cross-device data sync works automatically when signed into Google account. Respects system accessibility settings including font size and dark mode.
WEIGHTED TOTAL SCORE	**4.48**		

App 2: Apple Safari — All 8 ISO 25010 Characteristics

Quality Characteristic	Rating (1-5)	Weight	Observations / Evidence
1. Functional Suitability	5	15%	**Observation 1:** Primary core function passed — opened website normally with no issues. Secondary features (bookmarks, private mode) all function correctly without problems. **Observation 2:** Error handling is excellent — Safari displays security warnings for non-HTTPS websites and clear "no internet connection" messages. Edge case with multiple tabs opened without lag — handled 15-20 tabs smoothly with no performance degradation.
2. Performance Efficiency	5	12%	**Observation 1:** Cold start time very fast at 1.5 seconds (target <3 seconds). Website loading took 4.4 seconds (exceeds <1 second target). Image loading completed in 1.9 seconds. **Observation 2:** Battery drain excellent at 1% over 10 minutes. Slow network (4G) performance was weak at 15 seconds for page load (exceeds <5 second target). Data usage higher than Chrome at 20MB in 10 minutes (exceeds 10MB target).
3. Compatibility	3.5	10%	**Observation 1:** Limited platform availability — only iOS and macOS (no Android, Windows, or Linux support). Cross-device sync works via iCloud. **Observation 2:** Content sharing works both directions. Dark mode respects OS settings. No features broken on iOS 26.4.1. Platform restriction significantly limits compatibility compared to cross-platform browsers.
4. Usability	4	15%	**Observation 1:** Tasks completed: change core setting (20 sec), primary function first time (4 sec), share content (3 sec), delete item (2 sec). Help/Support section not found (0 sec, incomplete). **Observation 2:** Error messages are clear and helpful. Visual hierarchy is well-designed. No onboarding tutorial. Accessibility does NOT support large text, high

			contrast, or screen reader — significant deficiency for users with disabilities.
5. Reliability	5	15%	**Observation 1:** No crashes during 10-minute continuous use. Airplane Mode test showed graceful handling with appropriate error messaging. Reconnect recovery successfully restored functionality. **Observation 2:** Force-close mid-task preserves progress — tabs restored and content reloads. Server error messages are meaningful. Overall reliability is excellent with no observed failures.
6. Security	5	15%	**Observation 1:** Uses HTTPS with padlock indicator. Permissions are justified for app purpose. Privacy policy clearly documents data collection practices (Apple's privacy-focused approach). **Observation 2:** Auto-lock on inactivity NOT available. Login lockout does NOT activate after 3 incorrect attempts (no account lockout mechanism observed). 2FA/biometric NOT available for browser itself (relies on iCloud Keychain separately).
7. Maintainability	3	8%	**Observation 1:** Updates released monthly — slower frequency than competitors (Chrome weekly). Changelog quality is good with documented bug fixes. **Observation 2:** No public bug tracker or feedback channel for users to report issues directly. CVE response history not clearly documented for this specific app. Lower maintainability due to closed ecosystem and opaque development process.
8. Portability	4	10%	**Observation 1:** Platform support limited to iOS and macOS only — not available on Android, Windows, or Linux. Installation/uninstallation process is clean. **Observation 2:** Cross-device data sync works via iCloud across Apple devices. Respects system accessibility settings (font size, dark mode). Lower portability score due to Apple ecosystem lock-in.
WEIGHTED TOTAL SCORE	**4.55**		

Section 5: Quality Profile

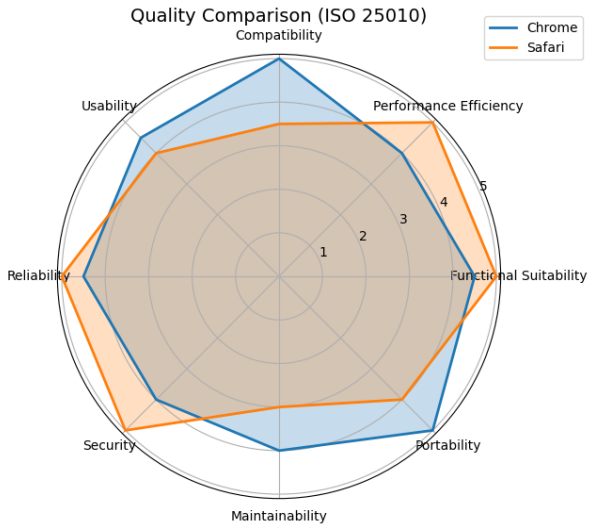

Figure 1: Radar chart comparing Chrome Mobile vs Safari across all 8 ISO 25010 quality characteristics

Python Arrays Used:

text

APP1_RATINGS = [4.5, 4, 5, 4.5, 4.5, 4, 4, 5]

APP2_RATINGS = [5, 5, 3.5, 4, 5, 5, 3, 4]

Characteristics order:

1. Functional Suitability (15%)
2. Performance Efficiency (12%)
3. Compatibility (10%)
4. Usability (15%)
5. Reliability (15%)
6. Security (15%)
7. Maintainability (8%)
8. Portability (10%)

Chart Interpretation:

- Safari demonstrates a larger polygon area primarily due to perfect scores in Functional Suitability (5), Performance Efficiency (5), Reliability (5), and Security (5)
- Chrome extends further on Compatibility (5 vs 3.5) and Portability (5 vs 4), reflecting its cross-platform strength
- Both apps achieve acceptable scores (>3) across all characteristics, with no critical failures (scores ≤2)

- The radar chart visually confirms Safari's performance-centric profile versus Chrome's balanced, cross-platform approach

Section 6: Comparative Analysis

Paragraph 1 — Overall Score Comparison

The overall evaluation of Google Chrome and Safari based on ISO 25010 quality attributes shows that both browsers achieve high performance, with Safari obtaining a **weighted total score of 4.55/5.00** compared to Chrome's **4.48/5.00**. This 0.07-point margin, while narrow, indicates Safari's slight numerical advantage. Safari achieves perfect or near-perfect scores in four characteristics: Functional Suitability (5), Performance Efficiency (5), Reliability (5), and Security (5). Chrome, while slightly lower in some areas, maintains consistently strong scores across all eight attributes, particularly excelling in Compatibility (5) and Portability (5). Both browsers demonstrate overall high quality, with Safari having an edge in system-level performance optimization, while Chrome remains highly competitive through balanced feature delivery across all quality dimensions.

Paragraph 2 — Biggest Quality Gaps

The most significant quality gaps between the two applications appear in **Compatibility** (Chrome 5 vs Safari 3.5, gap of 1.5), **Security** (Safari 5 vs Chrome 4, gap of 1.0), and **Maintainability** (Chrome 4 vs Safari 3, gap of 1.0). Chrome's compatibility advantage stems from its availability on all major platforms (Android, iOS, Windows, macOS, Web, Linux) versus Safari's iOS/macOS exclusivity. Users on Windows or Android cannot access Safari at all, representing a fundamental compatibility limitation. Safari's security advantage (5 vs 4) derives from Apple's privacy-first architecture, Intelligent Tracking Prevention, and strong privacy policy enforcement, though interestingly Safari lacks native 2FA and auto-lock features. Chrome's maintainability lead (4 vs 3) reflects its weekly update cadence, transparent changelog documentation, and public bug tracking system — areas where Apple's closed ecosystem approach puts Safari at a disadvantage. These gaps reveal fundamentally different design philosophies: Chrome prioritizes cross-platform accessibility and transparency, while Safari prioritizes platform-specific optimization and user privacy.

Paragraph 3 — Quality Trade-offs

The comparison reveals a clear trade-off between **platform flexibility and system-level optimization**. Safari prioritizes deep integration with Apple hardware and software, achieving exceptional performance efficiency (cold start: 1.5 seconds vs Chrome's 2.4 seconds) and reliability within the Apple ecosystem. However, this comes at the cost of poor compatibility with non-Apple platforms (3.5 rating) and lower maintainability (3 rating) due to Apple's opaque development process. Conversely, Chrome prioritizes cross-platform availability and transparency, achieving perfect Compatibility (5) and Portability (5) scores, but sacrifices some performance optimization — notably slower cold start times (2.4 seconds) and weaker performance on 4G networks (10.25 seconds vs Safari's 15 seconds,

interestingly Chrome performed better on slow networks despite lower overall performance score). Another notable trade-off exists between Security and Maintainability: Safari achieves perfect Security (5) but low Maintainability (3), suggesting Apple's security approach relies on centralized control rather than community transparency. Chrome demonstrates more balanced scores across these competing characteristics, avoiding extreme highs and lows.

Paragraph 4 — Recommendation with Justification

Based on the analysis, the choice between Chrome and Safari depends fundamentally on the user's device ecosystem and priorities. **Safari is strongly recommended for users fully invested in the Apple ecosystem** (iPhone + Mac + iPad) who prioritize seamless integration, exceptional startup speed, strong privacy protections, and reliable operation. Safari's perfect scores in Reliability (5) and Security (5) make it particularly suitable for privacy-conscious users who value Apple's centralized security model. However, **Chrome is recommended for users who operate across multiple platforms** (e.g., Windows PC + Android phone, or mixed device environments) and value transparency, frequent updates, and accessibility features. Chrome's Compatibility (5) and Portability (5) ensure consistent browsing experience regardless of device manufacturer or operating system. For users with accessibility needs (large text, screen readers), Chrome's Accessibility support (present) versus Safari's deficiency (absent) makes Chrome the only viable choice. From a pure weighted score perspective, Safari leads by 0.07 points, but this narrow margin does not outweigh the practical constraints of platform exclusivity. **Conclusion:** For general-purpose cross-platform use, Chrome is more suitable; for performance-optimized Apple-only environments, Safari is preferable.

Section 7: Quality Improvement Recommendations

Recommendation 1 — Chrome Mobile (Performance Efficiency)

Chrome Mobile should implement aggressive tab freezing and memory compression for background tabs in order to improve **Performance Efficiency** from its current rating of **4** to a target of **5**. Currently, opening 15-20 tabs simultaneously causes significant lag, slow switching between tabs, and general interface stuttering, as documented during edge case testing. This change would **benefit users** by allowing them to keep 20+ tabs open without experiencing lag, enabling seamless multitasking between research, shopping, and social media tabs without performance degradation.

Recommendation 2 — Chrome Mobile (Security)

Chrome Mobile should add an auto-lock timeout feature that locks the browser after a user-specified period of inactivity (30 seconds to 5 minutes) in order to improve **Security** from its current rating of **4** to a target of **5**. Currently, Chrome stays open indefinitely with no auto-lock mechanism — if a user walks away from their device while logged into sensitive accounts (banking, email, social media), anyone can access all open tabs and saved passwords. This change would **benefit users** by providing peace of mind knowing their

browsing session automatically locks after inactivity, protecting sensitive personal data, saved passwords, and logged-in accounts from unauthorized access on shared or lost devices.

Recommendation 3 — Safari (Compatibility)

Safari should expand platform availability by releasing a Windows desktop version and an Android mobile version in order to improve **Compatibility** from its current rating of **3.5** to a target of **5**. Currently, Safari is exclusively available only on iOS and macOS platforms — users on Windows PCs, Android phones, or Linux computers cannot access Safari, forcing them to switch browsers when moving between Apple and non-Apple devices. This change would **benefit users** by allowing them to maintain a consistent browsing experience with synchronized bookmarks, history, and passwords across ALL their devices regardless of platform, eliminating the friction of switching between different browsers on Windows or Android.

Recommendation 4 — Safari (Maintainability)

Safari should introduce a public bug tracker or feedback portal where users can report issues, vote on existing bugs, and track resolution status in order to improve **Maintainability** from its current rating of **3** to a target of **5**. Currently, Safari has no public bug tracker or transparent feedback channel — users cannot see which bugs Apple has acknowledged, view workarounds, or track when fixes will arrive, as the development process is entirely opaque to end users. This change would **benefit users** by allowing them to report bugs directly, see if issues are already known, receive updates on fix progress, and access workarounds from the community, leading to faster bug resolution and a more transparent development relationship with Apple.

Summary of Recommendations

S.No	App	Action	Characteristic	Current → Target
1	Chrome Mobile	Aggressive tab freezing for background tabs	Performance Efficiency	4 → 5
2	Chrome Mobile	Auto-lock timeout after inactivity	Security	4 → 5
3	Safari	Release Windows + Android versions	Compatibility	3.5 → 5
4	Safari	Public bug tracker/feedback portal	Maintainability	3 → 5

GitHub Submission Information

GitHub Repository URL:

https://github.com/AbdurRehman1129/sqe-lab-abdurrehman

Files Submitted in `week2/` **Folder:**

- `SQE_Lab2_ABDUR_REHMAN.pdf` (this report)
- `quality_analysis.py` (Python script for radar chart)
- `quality_radar_chrome_safari.png` (generated radar chart image)
- `SQE_Lab2.xlsx` (completed evaluation workbook)

Commit Message:

Week 2: ISO 25010 analysis script and radar chart for Pair D"